QF620 Stochastic Modelling in Finance — Session 6 Reference Cheatsheet

Valuation Framework and Stochastic Volatility Models: pricing vs reporting models, implied volatility, the volatility smile, fitting with Black–Scholes and Bachelier, displaced diffusion, stochastic volatility, Heston and SABR

Built from Session 6 slides (Tee Chyng Wen, SMU) · October 2026 · with symbol-by-symbol equation breakdowns

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Source: "Session 6: Valuation Framework and Stochastic Volatility Models" (Tee Chyng Wen, SMU, 20 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △**!** marks a slide ambiguity or error, with the corrected version.

How to read the equations: every displayed equation is followed by a shaded **where** box that names each symbol and component and says what it represents in that formula.

Exam context: the exam format is not stated on the slides; check the course outline. From this session

you need: the difference between pricing and reporting models; implied volatility and why the smile exists; why one-parameter models (Black–Scholes, Bachelier) cannot fit the smile; solving and pricing the displaced-diffusion model with the Black formula; the stochastic-volatility SDEs (lognormal vol, Heston, SABR) and their parameters; and how SABR is used (implied vol formula, then Black).

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
$S,\ F_t,\ F_0$	spot; forward price at t; today's forward	price	SABR and DD use forwards
$K,\ T,\ r$	strike, expiry, rate	—	
$C(K),\ P(K)$	market call/put price at strike K	price	[S4]
σ_K	implied volatility at strike K	number	[S4]
$\sigma_{\text{LN}},\ \sigma_{\text{N}}$	lognormal (Black–Scholes) / normal (Bachelier) volatility	number	[S8]
β	displaced-diffusion mix (DD), or CEV exponent (SABR)	number $\in [0,1]$	[S10, S17]
W_t^*	Brownian motion under the pricing measure	random	[S12]
ν	volatility of volatility	number	[S15]
$\sigma_t,\ \alpha_t$	stochastic volatility	random	[S15, S17]
V_t	instantaneous variance (Heston)	random	[S16]
$\kappa,\ \theta$	mean-reversion speed; long-run variance	number	Heston [S16]
ρ	correlation between price and vol Brownian motions	number $\in (-1,1)$	[S16–S17]
σ_{SABR}	SABR (Black) implied volatility	number	[S18]
`BlackCall`	Black (forward) call formula (F,K,σ,T)	function	[S11]

Notation trap: β means two different things: the displaced-diffusion weight on the lognormal part [S10] and the SABR backbone exponent F^β [S17]. In both, $\beta = 1$ is lognormal and $\beta = 0$ is normal. ν is vol-of-vol, V_t is a variance (Heston), σ_t and α_t are volatilities.

0. Topic map

```
PREREQUISITES
S5  Bachelier call, Black–Scholes call (d₁, d₂), solving SDEs with Itô;  S4 dW dW = dt
  |
  ▼
SESSION 6: VALUATION FRAMEWORK AND STOCHASTIC VOLATILITY MODELS
├─ REPORTING  [S2–S3]
|   ├─ Pricing models: price & hedge; realistic dynamics; accurate Greeks      [S2]
|   └─ Reporting models: convert market price ↔ implied vol, simple formula   [S3]
|
├─ SMILE  [S4–S7]
|   ├─ Implied vol σ_K: BS(S, K, r, σ_K, T) = market price                     [S4]
|   ├─ Flat before 1987; skewed after Black Monday                            [S5–S6]
|   └─ Steeper for short expiries; OTM puts in demand                          [S7]
|
├─ EXTENSIONS TO BLACK–SCHOLES  [S8–S9]
|   ├─ One parameter (σ_LN or σ_N): fit ATM only                               [S8]
|   └─ Google example: market smile lies between normal and lognormal          [S9]
|
├─ DISPLACED DIFFUSION  [S10–S14]
|   ├─ dF = σ[βF + (1–β)F₀]dW (Rubinstein 1983); solution via log             [S10]
|   ├─ Price with Black: Black(F₀/β, K + (1–β)F₀/β, σβ, T)                    [S11–S12]
|   ├─ Better fit for the right β, but not exact                               [S13]
|   └─ Real returns show volatility clustering ⇒ vol is random                [S14]
|
└─ STOCHASTIC VOLATILITY / SABR  [S15–S20]
    ├─ dσ = νσ dW^σ ⇒ σ_T = σ₀ exp(−½ν²T + νW_T^σ)                            [S15]
    ├─ Heston: dS = rS dt + √V S dW^S, dV = κ(θ–V)dt + ν√V dW^V, ρ              [S16]
    ├─ SABR: dF = αF^β dW^F, dα = να dW^α, ρ; σ_SABR → Black                   [S17]
```

```
   ├─ Hagan's σ_SABR formula                                          [S18]
   └─ DD lacks degrees of freedom; SABR fits the smile                [S19–S20]
  |
  ▼
LATER: calibration of SABR/Heston to market smiles; local volatility; exotic pricing
```

Dependency logic:

- Market prices are set by supply and demand. The Black–Scholes formula is used backwards as a *reporting* tool to turn each price into an implied volatility.
- If Black–Scholes were right, the implied vol would be flat across strikes. It is not (the smile), so models need more parameters.
- Displaced diffusion interpolates between normal and lognormal with one extra parameter β and is priced with the Black formula. It controls the slope (skew) but not the curvature.
- Making volatility itself random (vol-of-vol ν, correlation ρ) adds curvature and skew. SABR gives a closed-form implied vol, so it is fast to calibrate and widely used.

1 Concept entries

1.1 Pricing models vs reporting models [S2–S3]

Pricing models [S2] are used to price and risk-manage derivatives:

- their dynamics should match the modeller's intuition about how the underlying evolves over time;
- their Greeks should capture the derivatives' sensitivities accurately;
- each desk chooses its own models, with parameters (σ, β, …) **calibrated** to liquid option markets.

Reporting models [S3] are used by exchanges merely to *report* market prices (driven by supply and demand):

- it is more elegant to quote implied volatilities than prices, so the model converts price ↔ volatility;
- they make simplifying assumptions to get an analytically tractable formula (Black–Scholes, Bachelier);
- their parameters (implied vols) are shown on brokers' screens to communicate live prices.

Slide question [S3]: why quote vols rather than prices? *[Beyond slides]*: implied vol strips out the effect of spot, strike, expiry and rates, so options on different strikes, maturities and even underlyings can be compared on one scale; it changes much less than the price when spot moves; and it is what traders actually have views on.

1.2 Implied volatility [S4]

For each strike, the implied volatility is the σ that makes the Black–Scholes formula reproduce the observed market price:

$$C^{\text{mkt}}(K_i) = \text{BlackScholes}\big(S,\ K_i,\ r,\ \sigma_{K_i},\ T\big), \qquad i = 1, 2, 3, \ldots$$

where

$C^{\text{mkt}}(K_i)$	observed market price of the call (or put $P(K_i)$) with strike K_i.
S	spot price.
K_i	strike of the i-th option.
r	risk-free rate.
T	time to expiry.
σ_{K_i}	implied volatility for strike K_i: solved for numerically (the formula cannot be inverted in closed form). There is one per strike.

[Beyond slides] The solution is unique because the Black–Scholes price is strictly increasing in σ (vega $= S\phi(d_1)\sqrt{T} > 0$). By put–call parity, a call and a put with the same strike and expiry have the same implied vol.

1.3 The volatility smile [S5–S7]

- Black–Scholes assumes volatility is a **single constant**, independent of strike and expiry, so implied vol plotted against strike should be a horizontal line [S5–S6].
- Before the October 1987 crash (Black Monday) the S&P 500 index smile was roughly flat. Afterwards a distinct **smile/skew** appeared across many markets: the market learned that a 20%+ drop in a short time was possible [S5–S6].

- In liquid markets, prices come from supply and demand, not a model. Higher implied vols for low strikes (OTM puts) reflect demand for **protection against large downward moves** [S6].
- The smile is **steepest for short expiries** and flatter for long ones [S7].
- The market trades mostly OTM and ATM options; ITM options are less liquid. For equity indices this means heavy demand for OTM puts [S7].

[Beyond slides] Equivalent view: the market's risk-neutral distribution has a fatter left tail (negative skew, excess kurtosis) than the lognormal.

1.4 Fitting market prices with one-parameter models [S8–S9]

With Black–Scholes or Bachelier the only free parameter is the volatility, so it is chosen to fit the most important option, the **ATM** one, and the same value is used for every strike [S8]:

$$\text{model price}(K_i) = \text{BlackScholes}(S, K_i, \sigma_{\text{LN}}, T) \quad \text{or} \quad \text{Bachelier}(S, K_i, \sigma_{\text{N}}, T).$$

where

σ_{LN}	lognormal (percentage) volatility: one number for all strikes.
σ_{N}	normal (price-unit) volatility: one number for all strikes. *[Beyond slides]* To match ATM, $\sigma_{\text{N}} \approx \sigma_{\text{LN}} S$.

Google example [S9]: options on 2013-08-30 expiring 2015-01-17 ($T \approx 1.38$), spot 846.9, ATM vol ≈ 0.26. Converting each model's prices to Black–Scholes implied vols:

- **Lognormal model:** flat at 0.26 for all strikes.
- **Normal model:** downward-sloping (higher implied vol at low strikes). *[Beyond slides]* With $\sigma_{\text{N}} = 0.26 \times 846.9$: about 0.37 at $K = 400$, 0.26 ATM, 0.20 at $K = 1400$.
- **Market:** between the two (steeper than lognormal, flatter than normal), with some upturn at high strikes.

1.5 Displaced-diffusion model: SDE and solution [S10]

Rubinstein (1983):

$$\mathrm{d}F_t = \sigma\big[\beta F_t + (1-\beta)F_0\big]\,\mathrm{d}W_t, \qquad \beta \in [0,1].$$

where

F_t	forward price (driftless under the pricing measure).
F_0	today's forward: a fixed constant.
βF_t	lognormal (geometric) part: noise proportional to the level.
$(1-\beta)F_0$	normal (arithmetic) part: noise of fixed size.
β	mixing weight: $\beta = 1$ gives Black ($\mathrm{d}F = \sigma F\,\mathrm{d}W$); $\beta = 0$ gives Bachelier with $\sigma_{\text{N}} = \sigma F_0$.
σ	volatility; at $t = 0$ the local volatility is σF_0, the same as Black's.

Itô on $X_t = f(F_t) = \log\big[\beta F_t + (1-\beta)F_0\big]$ with $f' = \frac{\beta}{\beta F + (1-\beta)F_0}$, $f'' = -\frac{\beta^2}{[\beta F + (1-\beta)F_0]^2}$ (working *[Beyond slides]*):

$$\mathrm{d}X_t = -\tfrac{1}{2}\beta^2\sigma^2\,\mathrm{d}t + \beta\sigma\,\mathrm{d}W_t \quad \Longrightarrow \quad F_T = \frac{F_0}{\beta}\exp\Big[-\frac{\beta^2\sigma^2 T}{2} + \beta\sigma W_T\Big] - \frac{1-\beta}{\beta}F_0.$$

where

$\frac{1}{2}\sigma_t^2 f''$	with $\sigma_t = \sigma[\beta F + (1-\beta)F_0]$: gives $-\frac{1}{2}\beta^2\sigma^2$ (no drift term since $\mathrm{d}F$ has none).
$\sigma_t f'$	$= \beta\sigma$: constant, so X is an arithmetic BM.
$X_0 = \log F_0$	since $\beta F_0 + (1-\beta)F_0 = F_0$.
$\frac{F_0}{\beta}e^{(\cdot)}$	exponentiate X_T and solve $\beta F_T + (1-\beta)F_0 = F_0 e^{(\cdot)}$ for F_T.
$-\frac{1-\beta}{\beta}F_0$	the displacement (shift): F_T is a **shifted lognormal**, bounded below by $-\frac{1-\beta}{\beta}F_0$.

The SDE combines a geometric and an arithmetic Brownian motion [S10].

1.6 Pricing displaced diffusion with the Black formula [S11–S12]

Slide question [S11]: compare

$$\text{Black: } F_T = F_0 e^{-\frac{\sigma^2 T}{2}+\sigma W_T}, \qquad \text{DD: } F_T = \frac{F_0}{\beta} e^{-\frac{\beta^2\sigma^2 T}{2}+\beta\sigma W_T} - \frac{1-\beta}{\beta}F_0.$$

where

Black F_T	lognormal forward with start F_0 and vol σ.
DD F_T	a Black forward with start F_0/β and vol $\beta\sigma$, shifted down by $\frac{1-\beta}{\beta}F_0$.

So yes, `BlackCall` can price it. Moving the shift onto the strike:

$$(F_T-K)^+ = \Big(\underbrace{\tfrac{F_0}{\beta}e^{-\frac{1}{2}(\beta\sigma)^2 T+\beta\sigma W_T}}_{\text{Black forward}} - \Big[K+\tfrac{1-\beta}{\beta}F_0\Big]\Big)^+ \quad\Longrightarrow\quad \text{DD call} = \text{Black}\Big(\frac{F_0}{\beta},\ K+\frac{1-\beta}{\beta}F_0,\ \sigma\beta,\ T\Big).$$

where

$\frac{F_0}{\beta}$	"forward" input to Black.
$K+\frac{1-\beta}{\beta}F_0$	"strike" input: the real strike plus the displacement.
$\sigma\beta$	"volatility" input.
$\text{Black}(F,K,\sigma,T)$	$D\,[F\Phi(d_1)-K\Phi(d_2)]$, $d_{1,2}=\frac{\log(F/K)\pm\frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$, D = discount factor.

[S12]: the market smile lies **between normal and lognormal**, and displaced diffusion has features of both. It is written with W_t^*, a Brownian motion under the pricing (forward) measure, so F_t is driftless.

1.7 Fitting with displaced diffusion; why vol must be random [S13–S14]

- Choosing β (0.2–0.8 shown) gives a **closer fit** than either pure model, since β tilts the skew between flat (lognormal) and steep (normal) [S13].
- The fit is still **not accurate enough**: displaced diffusion produces a monotone skew and cannot match the curvature (the upturn at high strikes) [S13].
- **Which returns are real?** [S14] (answer *[Beyond slides]*): the left series. It shows **volatility clustering**: calm periods and turbulent periods (e.g. 2008) with big spikes and fat tails. The right series has constant variance (i.i.d. normal), which is what Black–Scholes assumes. So volatility should itself be random.

1.8 Stochastic volatility [S15]

Daily log returns show volatility is stochastic, so give it its own SDE. Simplest: a driftless lognormal process:

$$\mathrm{d}\sigma_t = \nu\,\sigma_t\,\mathrm{d}W_t^{\sigma}.$$

where

σ_t	volatility at time t: now a random process.
ν	**volatility of volatility** (vol-of-vol): how much σ itself fluctuates.
W_t^{σ}	Brownian motion driving the volatility (may be correlated with the price's).
driftless	$\mathbb{E}[\sigma_t]=\sigma_0$.

Solved like geometric Brownian motion (Itô on $\log\sigma_t$):

$$\sigma_T = \sigma_0 \exp\Big[-\frac{1}{2}\nu^2 T+\nu W_T^{\sigma}\Big] = \sigma_0\exp\Big[-\frac{1}{2}\nu^2 T+\nu\sqrt{T}\,N(0,1)\Big].$$

where

σ_0	today's volatility.
$-\frac{1}{2}\nu^2 T$	Itô convexity correction, keeping $\mathbb{E}[\sigma_T]=\sigma_0$.
$\nu\sqrt{T}\,N(0,1)$	normal shock to log-vol: σ_T is lognormal and stays positive.

1.9 Heston model [S16]

Steven Heston (1993):

$$\begin{cases} \mathrm{d}S_t = rS_t\,\mathrm{d}t + \sqrt{V_t}\,S_t\,\mathrm{d}W_t^S \\ \mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \nu\sqrt{V_t}\,\mathrm{d}W_t^V \end{cases} \qquad \mathrm{d}W_t^S\,\mathrm{d}W_t^V = \rho\,\mathrm{d}t.$$

where

S_t	stock price; drift r under the risk-neutral measure.
V_t	instantaneous **variance**; $\sqrt{V_t}$ is the volatility.
κ	speed at which variance mean-reverts.
θ	long-run variance level.
ν	vol-of-vol (here, volatility of variance per $\sqrt{V}$).
$\nu\sqrt{V_t}\,\mathrm{d}W^V$	square-root diffusion (CIR): noise shrinks as $V \to 0$, keeping $V \geq 0$.
ρ	correlation of stock and variance shocks; $\rho < 0$ for equities produces the downward skew (leverage effect) *[Beyond slides]*.

- Variance follows a **mean-reverting square-root diffusion** [S16].
- Vanilla European prices need a one-dimensional integral evaluated numerically (Fourier, characteristic function) [S16].
- Popular with equity desks [S16].
- *[Beyond slides]* Feller condition $2\kappa\theta > \nu^2$ keeps V_t strictly positive.

1.10 SABR model [S17]

Stochastic alpha–beta–rho, Patrick Hagan (2002):

$$\begin{cases} \mathrm{d}F_t = \alpha_t\,F_t^{\beta}\,\mathrm{d}W_t^F \\ \mathrm{d}\alpha_t = \nu\,\alpha_t\,\mathrm{d}W_t^{\alpha} \end{cases} \qquad \mathrm{d}W_t^F\,\mathrm{d}W_t^{\alpha} = \rho\,\mathrm{d}t.$$

where

F_t	forward price (driftless).
α_t	stochastic volatility level; $\alpha_0 = \alpha$ is today's value.
F_t^{β}	backbone (CEV): $\beta = 1$ lognormal, $\beta = 0$ normal, in between a mix.
ν	vol-of-vol: controls the smile's **curvature**.
ρ	correlation of forward and vol shocks: controls the **skew** (tilt).
$\mathrm{d}\alpha = \nu\alpha\,\mathrm{d}W$	zero-drift lognormal volatility, as in §1.8.

Use [S17]: Hagan derived an analytic approximation σ_{SABR} for the implied volatility. To price a vanilla: compute $\sigma_{\mathrm{SABR}}(F_0, K, \dots)$ and plug it into the Black formula. Much faster than Heston; used across many asset classes.

1.11 Hagan's SABR implied volatility [S18]

$$\sigma_{\mathrm{SABR}}(F_0,K,\alpha,\beta,\rho,\nu) = \frac{\alpha}{(F_0K)^{(1-\beta)/2}\left\{1 + \frac{(1-\beta)^2}{24}\log^2\frac{F_0}{K} + \frac{(1-\beta)^4}{1920}\log^4\frac{F_0}{K} + \cdots\right\}} \times \frac{z}{x(z)} \times \left\{1 + \left[\frac{(1-\beta)^2}{24}\frac{\alpha^2}{(F_0K)^{1-\beta}} + \frac{1}{4}\frac{\rho\beta\nu\alpha}{(F_0K)^{(1-\beta)/2}} + \frac{2-3\rho^2}{24}\nu^2\right]T + \cdots\right\}$$

where

$\frac{\alpha}{(F_0K)^{(1-\beta)/2}}$	converts the CEV vol αF^{β} into a lognormal vol at the geometric-mean level $\sqrt{F_0K}$.
$\{1 + \frac{(1-\beta)^2}{24}\log^2\cdots\}$	correction for the backbone away from ATM; vanishes when $\beta = 1$.
$\frac{z}{x(z)}$	the stochastic-vol smile factor: depends on ν and ρ; equals 1 at the money.
$[\cdots]T$	small time-dependent correction (three terms: backbone, correlation, vol-of-vol).
log	natural log.

with

$$z = \frac{\nu}{\alpha}(F_0K)^{(1-\beta)/2}\log\left(\frac{F_0}{K}\right), \qquad x(z) = \log\left[\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}\right].$$

where

z moneyness $\log(F_0/K)$ scaled by vol-of-vol over vol.

$x(z)$ nonlinear transform carrying the skew (ρ) and curvature (ν).

- *[Beyond slides]* ATM ($K = F_0$): $z \to 0$ and $z/x(z) \to 1$, giving $\sigma_{\text{ATM}} \approx \frac{\alpha}{F_0^{1-\beta}}\{1 + [\dots]T\}$. Code must handle this limit separately.
- *[Beyond slides]* $\beta = 1$, $\nu \to 0$: $\sigma_{\text{SABR}} = \alpha$, flat (Black).
- *[Beyond slides]* Practice: fix β (often 0.5 or by asset class), calibrate α, ρ, ν to the market smile for each expiry.
- Code is in the course's separate Jupyter notebook [S18].

1.12 Fit comparison [S19–S20]

- Displaced diffusion (best β) has **too few degrees of freedom**: it misses the low-strike steepness and the high-strike upturn [S19].
- SABR fits the implied volatility smile well and is quick to compute, hence popular [S20].

1.13 Implementation *[Beyond slides]* (tested)

```
import numpy as np
from scipy.stats import norm
from scipy.optimize import brentq
def black_call(F, K, sigma, T, D=1.0):
    d1 = (np.log(F/K) + 0.5*sigma**2*T) / (sigma*np.sqrt(T))
    return D*(F*norm.cdf(d1) - K*norm.cdf(d1 - sigma*np.sqrt(T)))
def bachelier_call(F, K, sigma, T, D=1.0):
    d = (F - K) / (sigma*np.sqrt(T))
    return D*((F - K)*norm.cdf(d) + sigma*np.sqrt(T)*norm.pdf(d))
def implied_vol(price, F, K, T, D=1.0):
    return brentq(lambda s: black_call(F, K, s, T, D) - price, 1e-6, 5.0)
def dd_call(F, K, sigma, beta, T, D=1.0):
    """Displaced diffusion dF = sigma [beta F + (1-beta) F0] dW, priced with Black."""
    return black_call(F/beta, K + (1-beta)/beta*F, sigma*beta, T, D)
def sabr_vol(F, K, T, alpha, beta, rho, nu):
    """Hagan et al. (2002) lognormal implied vol."""
    B = lambda fk: 1 + ((1-beta)**2/24*alpha**2/fk**(1-beta)
                        + 0.25*rho*beta*nu*alpha/fk**((1-beta)/2) + (2-3*rho**2)/24*nu**2)*T
    if abs(F - K) < 1e-12:                                  # ATM: z/x(z) -> 1
        return alpha/F**(1-beta)*B(F*F)
    fk, lfk = F*K, np.log(F/K)
    z = nu/alpha*fk**((1-beta)/2)*lfk
    xz = np.log((np.sqrt(1 - 2*rho*z + z**2) + z - rho) / (1 - rho))
    A = alpha / (fk**((1-beta)/2)*(1 + (1-beta)**2/24*lfk**2 + (1-beta)**4/1920*lfk**4))
    return A*(z/xz)*B(fk)
F, T = 846.9, 1.38                                          # Google example
for K in (400, 846.9, 1400):                                # normal model -> skew in Black vols
    print(K, round(implied_vol(bachelier_call(F, K, 0.26*F, T), F, K, T), 4))  # 0.3725 0.261 0.2006
print(round(dd_call(F, 900, 0.26, 0.4, T), 3))              # 80.009 (Monte Carlo: 79.963 +/- 0.05)
print(round(sabr_vol(F, 1200, T, 0.26, 1.0, 0.0, 1e-8), 6))  # 0.26: beta=1, nu->0 is flat Black
```

2 Comparison tables

2.1 Pricing vs reporting models

	Pricing model [S2]	Reporting model [S3]
Purpose	price and hedge derivatives	quote market prices as vols
Requirement	realistic dynamics, accurate Greeks	simple, analytic formula
Parameters	calibrated to liquid options	implied vols on screens
Examples	SABR, Heston, displaced diffusion	Black–Scholes, Bachelier

2.2 Models and the smile

Model	SDE	Free params	Smile in BS implied vol
Black–Scholes	$\mathrm{d}F = \sigma F\,\mathrm{d}W$	σ	flat
Bachelier	$\mathrm{d}F = \sigma_{\mathrm{N}}\,\mathrm{d}W$	σ_{N}	fixed downward skew
Displaced diffusion	$\mathrm{d}F = \sigma[\beta F + (1-\beta)F_0]\mathrm{d}W$	σ, β	adjustable skew, no curvature
Heston	see §1.9	$V_0, \kappa, \theta, \nu, \rho$	skew and curvature; numerical integral
SABR	see §1.10	α, β, ρ, ν	skew (ρ, β) and curvature (ν); closed-form vol

2.3 What each SABR parameter does *[Beyond slides]*

α	overall level of volatility (mainly ATM vol)
β	backbone: how ATM vol moves with F; usually fixed
ρ	skew: $\rho < 0$ raises low-strike vols
ν	curvature: higher ν lifts both wings

2.4 Heston vs SABR

	Heston [S16]	SABR [S17–S18]
Random quantity	variance V_t, mean-reverting	volatility α_t, driftless lognormal
Underlying	spot with drift r	forward, driftless, CEV F^β
Vanilla pricing	1-d numerical integral	closed-form implied vol + Black
Speed	slower	fast
Typical use	equity desks	all asset classes, esp. rates

3 Master formula sheet

Each line repeats the key symbols so the sheet can be used on its own.

Implied volatility [S4]

- $C^{\mathrm{mkt}}(K) = \mathrm{BS}(S, K, r, \sigma_K, T)$, solved for σ_K; unique since vega > 0.
  where σ_K = implied vol at strike K.

Black and Bachelier (forward form)

- Black: $D[F\Phi(d_1) - K\Phi(d_2)]$, $d_{1,2} = \frac{\log(F/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$; Bachelier: $D[(F-K)\Phi(d) + \sigma_{\mathrm{N}}\sqrt{T}\phi(d)]$, $d = \frac{F-K}{\sigma_{\mathrm{N}}\sqrt{T}}$.
  where D = discount factor; F = forward.

Displaced diffusion [S10–S12]

- $\mathrm{d}F = \sigma[\beta F + (1-\beta)F_0]\mathrm{d}W$; $F_T = \frac{F_0}{\beta} e^{-\frac{1}{2}\beta^2\sigma^2 T + \beta\sigma W_T} - \frac{1-\beta}{\beta}F_0$.
  where $\beta = 1$ lognormal; $\beta = 0$ normal.
- $\mathrm{Call} = \mathrm{Black}\big(\frac{F_0}{\beta},\ K + \frac{1-\beta}{\beta}F_0,\ \sigma\beta,\ T\big)$.
  where forward, strike, vol inputs to Black.

Stochastic volatility [S15–S18]

- $\mathrm{d}\sigma = \nu\sigma\,\mathrm{d}W^\sigma \Rightarrow \sigma_T = \sigma_0 e^{-\frac{1}{2}\nu^2 T + \nu W_T^\sigma}$.
  where ν = vol-of-vol.
- Heston: $\mathrm{d}S = rS\mathrm{d}t + \sqrt{V}S\mathrm{d}W^S$, $\mathrm{d}V = \kappa(\theta - V)\mathrm{d}t + \nu\sqrt{V}\mathrm{d}W^V$, $\mathrm{d}W^S\mathrm{d}W^V = \rho\mathrm{d}t$.
  where V = variance; κ = speed; θ = long-run variance.

- SABR: $\mathrm{d}F = \alpha F^{\beta}\mathrm{d}W^{F}$, $\mathrm{d}\alpha = \nu\alpha\mathrm{d}W^{\alpha}$, $\mathrm{d}W^{F}\mathrm{d}W^{\alpha} = \rho\mathrm{d}t$; price $= \mathrm{Black}(F_0, K, \sigma_{\mathrm{SABR}}, T)$.
  where σ_{SABR} from Hagan's formula (§1.11); $z = \frac{\nu}{\alpha}(F_0K)^{\frac{1-\beta}{2}}\log\frac{F_0}{K}$.

4 Glossary

- **ATM / ITM / OTM:** at / in / out of the money [S7].
- **Backbone:** how ATM vol moves with the forward (SABR β) *[Beyond slides]*.
- **Calibration:** choosing model parameters to fit market prices [S2].
- **Displaced diffusion (shifted lognormal):** $\mathrm{d}F = \sigma[\beta F + (1-\beta)F_0]\mathrm{d}W$ [S10].
- **Heston model:** stock with mean-reverting square-root variance [S16].
- **Implied volatility:** the σ that makes Black–Scholes match a market price [S4].
- **Leverage effect:** negative correlation between price and volatility moves *[Beyond slides]*.
- **Pricing model:** used to price and hedge [S2].
- **Reporting model:** converts prices to implied vols for quoting [S3].
- **SABR:** stochastic alpha–beta–rho model with closed-form implied vol [S17].
- **Skew / smile:** implied vol varying with strike [S5].
- **Stochastic volatility:** volatility follows its own SDE [S15].
- **Vol-of-vol ν:** volatility of the volatility process [S15].
- **Volatility clustering:** large moves follow large moves; calm follows calm [S14].

5 Self-test

Questions

Q1 ([S2–S3]). Contrast pricing and reporting models. Why are implied vols quoted rather than prices?

Q2 ([S4]). Define implied volatility. Why is it unique?

Q3 ([S5–S7]). Describe the equity smile, when it appeared and why.

Q4 ([S8–S9]). Why can't Black–Scholes or Bachelier fit the market smile? What shape does Bachelier produce in Black implied vol?

Q5 ([S10]). Solve $\mathrm{d}F = \sigma[\beta F + (1-\beta)F_0]\mathrm{d}W$. What are the limits $\beta = 1$ and $\beta \to 0$?

Q6 ([S11–S12]). Price a DD call with `BlackCall`. Give the inputs.

Q7 ([S13–S14]). Why is DD not enough, and what evidence shows vol is random?

Q8 ([S15]). Solve $\mathrm{d}\sigma = \nu\sigma\,\mathrm{d}W$ and find $\mathbb{E}[\sigma_T]$.

Q9 ([S16]). Write the Heston SDEs and explain each parameter.

Q10 ([S17–S18]). Write the SABR SDEs, state what ρ and ν control, and explain how SABR prices a vanilla.

Solutions

S1. See §1.1 and table 2.1. Vols normalise for spot, strike, expiry and rates, so they are comparable and more stable.

S2. The σ_K with $\mathrm{BS}(S, K, r, \sigma_K, T) = C^{\mathrm{mkt}}(K)$. Unique because price is strictly increasing in σ.

S3. Low strikes have higher implied vol (downward skew), steeper for short expiries; appeared after the 1987 crash; demand for OTM put protection.

S4. One parameter fits only the ATM option. Bachelier gives Black vols falling with strike (0.37 / 0.26 / 0.20 at $K = 400$ / 846.9 / 1400 in the Google example).

S5. Itô on $\log[\beta F + (1-\beta)F_0]$: $F_T = \frac{F_0}{\beta}e^{-\frac{1}{2}\beta^2\sigma^2T + \beta\sigma W_T} - \frac{1-\beta}{\beta}F_0$. $\beta = 1$: Black. $\beta \to 0$: $F_T \to F_0 + \sigma F_0 W_T$, Bachelier with $\sigma_{\mathrm{N}} = \sigma F_0$.

S6. $\mathrm{Black}\big(F_0/\beta,\ K + \frac{1-\beta}{\beta}F_0,\ \sigma\beta,\ T\big)$.

S7. One extra parameter only tilts the skew; it cannot produce curvature. Real returns show volatility clustering and fat tails.

S8. $\sigma_T = \sigma_0 e^{-\frac{1}{2}\nu^2T + \nu W_T}$; $\mathbb{E}[\sigma_T] = \sigma_0$.

S9. See §1.9.

S10. See §1.10. ρ = skew; ν = curvature. Compute $\sigma_{\text{SABR}}(F_0, K)$ with Hagan's formula, then use the Black formula.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	Pricing models	1.1
S3	Reporting models	1.1
S4	Implied volatility per strike	1.2
S5	Smile before and after 1987	1.3
S6	Smile background; demand for protection	1.3
S7	Smile by expiry; OTM/ATM/ITM liquidity	1.3
S8	Fitting with one volatility parameter	1.4
S9	Google example: market vs normal vs lognormal	1.4
S10	Displaced diffusion SDE and solution	1.5
S11	Pricing DD with BlackCall (question)	1.6
S12	DD between normal and lognormal; Black inputs	1.6
S13	DD fit for several β	1.7
S14	Real vs simulated returns	1.7
S15	Stochastic volatility; vol-of-vol	1.8
S16	Heston model	1.9
S17	SABR model	1.10
S18	Hagan's SABR implied vol formula	1.11
S19	DD fit (insufficient degrees of freedom)	1.12
S20	SABR fit	1.12

Appendix B. Flag summary (△!)

1. [S10, S17] β is used for two different parameters (DD weight and SABR exponent).
2. [S11–S12] The `BlackCall` signature omits the discount factor; multiply by $D = e^{-rT}$ when F is a forward.
3. [S16] Heston's ν is the volatility of *variance*, unlike SABR's ν (volatility of volatility).
4. [S18] The formula has a removable singularity at $K = F_0$ ($z/x(z) \to 1$); implementations must treat ATM separately.
5. [S3, S10, S11, S13, S14] Questions posed on the slides are answered here with my own working.